

Roll No. :S006

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Class :SYIT

Batch :01

Date of Assignment :05/08/2026

Date/Time of Submission :05/08/2026

PRACTICAL 4

```
import numpy as np
```

a. Write a python code to create Numpy Array.

```
arr = np.array([10, 20, 30, 40, 50])  
  
print("NumPy Array:",arr)
```

NumPy Array: [10 20 30 40 50]

b. Write a python code to demonstrate basic operations on single array.

```
print("Original Array:", arr)  
print("Addition (+10):", arr + 10)  
print("Multiplication (*2):", arr * 2)
```

Original Array: [10 20 30 40 50]
Addition (+10): [20 30 40 50 60]
Subtraction (-5): [5 15 25 35 45]
Multiplication (*2): [20 40 60 80 100]
Division (/2): [5. 10. 15. 20. 25.]

c. Write a python code to create array with 10 elements and slice element from 1st to 5th element. **bold text**

```
arr = np.array([10, 20, 30, 40, 50, 60, 70, 80, 90, 100])  
  
print("Original Array:", arr)  
  
print("Sliced Array:", arr[0:5])
```

Original Array: [10 20 30 40 50 60 70 80 90 100]
Sliced Array: [10 20 30 40 50]

d. Write a python code to sort an array alphabetically.

```
name= np.array(["Banana", "Apple", "Orange", "Mango", "Grapes"])

print("Original Array:", name)
print("Sorted Array:",np.sort(name) )
```

```
Original Array: ['Banana' 'Apple' 'Orange' 'Mango' 'Grapes']
Sorted Array: ['Apple' 'Banana' 'Grapes' 'Mango' 'Orange']
```

e. Write a python code to create a filter array that will return maximum values from an array.

bold text

```
arr = np.array([12, 45, 67, 89, 34, 89, 23])
print("Original Array:", arr)
print("Maximum Value:", np.max(arr))
print("Filtered Array:", (arr[arr>50]))
```

```
Original Array: [12 45 67 89 34 89 23]
Maximum Value: 89
Filtered Array: [67 89 89]
```